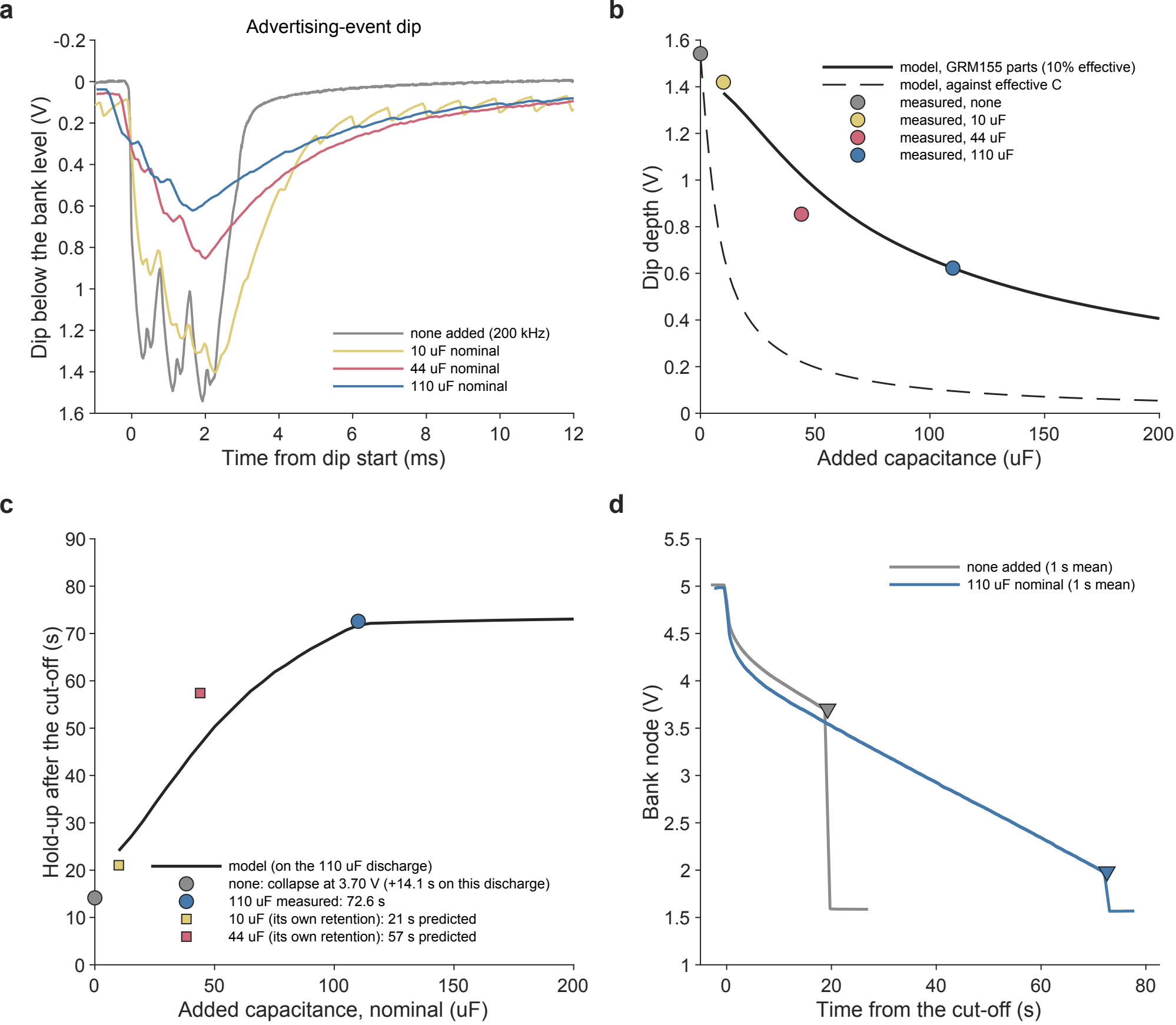

a
Advertising-event dip
Dip below the bank level (V)
Time from dip start (ms)
none added (200 kHz)
10 uF nominal
44 uF nominal
110 uF nominal
b
Dip depth (V)
Added capacitance (uF)
model, GRM155 parts (10% effective)
model, against effective C
measured, none
measured, 10 uF
measured, 44 uF
measured, 110 uF
c
Hold-up after the cut-off (s)
Added capacitance, nominal (uF)
model (on the 110 uF discharge)
none: collapse at 3.70 V (+14.1 s on this discharge)
110 uF measured: 72.6 s
10 uF (its own retention): 21 s predicted
44 uF (its own retention): 57 s predicted
d
Bank node (V)
Time from the cut-off (s)
none added (1 s mean)
110 uF nominal (1 s mean)

Decoupling capacitor tests: radio dips and hold-up (device not connected, advertising)

Parts: Murata GRM155R60J226ME11 = 22 uF, X5R, 6.3 V, 0402, +/-20% (2 x = 44 uF, 5 x = 110 uF nominal);
10 uF X7R 0603, 6.3 V. The file names already give the right rating (6.3 V).

1. Radio event with no added capacitor (DC_5.0V_CapBank_Test_2.bin, 200 kHz)
Advertising events every ~190 ms: three bursts over ~3 ms, dip 1.54 V; charge 11.7 uC per event
(i = x/Rs + Cn dx/dt, Rs = 322 ohm, Cn = 0.71 uF). Hold-up 19.3 s; collapse at mean 3.70 V.

2. Effective capacitance of the added parts (model dip matched to the measured dip)
1 x 10 uF X7R 0603, 6.3 V depth 1.42 V at 3.97 V -> 0.7 uF effective (7% of nominal)
2 x Murata GRM155R60J226ME11 (22 uF X5R 0402, 6.3 V) depth 0.85 V at 3.02 V -> 6.7 uF effective (15% of nominal)
5 x Murata GRM155R60J226ME11 (22 uF X5R 0402, 6.3 V) depth 0.62 V at 3.82 V -> 11.5 uF effective (10% of nominal)
The 0402 22 uF parts keep only about a quarter of their capacitance at 3-4 V (DC bias), as expected
for a 6.3 V 0402 X5R part.

3. Hold-up
110 uF nominal: 72.6 s, collapse at mean 1.98 V (none: 19.3 s, 3.70 V). On the same discharge,
reaching 3.70 V took 14.1 s, so the capacitor added about 58 s for the same bank.
Collapse rule from the two runs: Vcol = 0.81 V + 1.88 x dip.

4. Return for 10-200 uF of GRM155R60J226ME11 (see decoupling_tests_curve.csv)
10 uF nominal (1.0 uF effective): dip 1.37 V, collapse at 3.40 V, hold-up 24.1 s (+10.0 s)
20 uF nominal (2.1 uF effective): dip 1.27 V, collapse at 3.21 V, hold-up 30.3 s (+16.1 s)
45 uF nominal (4.7 uF effective): dip 1.01 V, collapse at 2.71 V, hold-up 47.2 s (+33.1 s)
65 uF nominal (6.8 uF effective): dip 0.85 V, collapse at 2.41 V, hold-up 57.8 s (+43.7 s)
110 uF nominal (11.5 uF effective): dip 0.62 V, collapse at 1.98 V, hold-up 71.7 s (+57.6 s)
150 uF nominal (15.7 uF effective): dip 0.50 V, collapse at 1.76 V, hold-up 72.6 s (+58.5 s)
200 uF nominal (21.0 uF effective): dip 0.41 V, collapse at 1.58 V, hold-up 73.0 s (+58.9 s)

5. Conclusion
Most of the gain comes in the first ~50 uF nominal (+36 s); 100 uF gives +55 s and 200 uF +59 s.
Past ~100 uF the device runs down close to its floor voltage and extra capacitance adds little.
The 0402 6.3 V parts are inefficient at this voltage: a 10 V (or 16 V) 0603/0805 X5R/X7R part keeps far
more capacitance, so fewer parts give the same effective value; a polymer capacitor barely derates.
Caveats: two hold-up runs only (no hold-up with 10 or 44 uF); the bank discharge differs between runs,
so hold-up times are compared on one discharge (the 110 uF run); Rs and Cn are taken as fixed.